Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | sahaj-bhadja | sahajbhadja | sahaj-bhadja

Summary

Software engineer with 2+ years building **systems-oriented Python services**, **data processing pipelines**, and performance-sensitive applications, with hands-on experience in **C++**, Linux workflows, testing, and cross-functional delivery of reliable production software.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*

- Built **Python-first platform services** for 500+ users, reducing request latency from **1.2s to 350ms** by optimizing execution paths, tuning SQL access patterns, and improving responsiveness for analytics workflows.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures while improving runtime observability, release confidence, and robustness across recurring production data-processing workflows.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*

- Developed **full-stack features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable backend services and responsive interfaces while improving delivery predictability through clearer ownership and sequencing.
- Built **AI-assisted grading workflows** with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting production-scale operations with fault-aware batch execution and CI/CD-backed release practices.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*

- Built **large-scale experimentation pipelines** across 100,000+ trajectory records, enabling reproducible algorithm evaluation and high-confidence benchmarking while improving signal quality for debugging model-driven decision workflows.
- Produced **technical reports and test scenarios** translating findings into actionable guidance, improving cross-team alignment, reducing handoff ambiguity, and strengthening engineering quality as prototypes matured toward production readiness.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*

- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations, resilient data movement, and stable delivery across production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence in frequent releases across coordinated service components.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*

- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening real-time responsiveness for high-volume user-facing processes.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation, troubleshooting, and handoffs.

Skills

Languages: **Python**, **C++**, Java, TypeScript, JavaScript, SQL, Bash
Systems & Backend: FastAPI, Flask, Node.js, REST APIs, ETL pipelines, WebSockets, distributed workflows
Data & Compute: PostgreSQL, MySQL, SQL Server, MongoDB, query tuning, batch processing, analytics pipelines
Cloud & Tooling: AWS, Docker, Kubernetes, Terraform basics, Jenkins, Git/GitHub, Linux/UNIX
Engineering: automated testing, CI/CD, Agile/Scrum, technical documentation, debugging, cross-functional collaboration

Projects

CPU-Scheduling Simulator Live | *C++, React, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, and MLQ scheduling** in C++, pairing algorithmic correctness with visualization tooling to improve debugging efficiency and analyze performance across preemptive execution scenarios.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production ETL workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring and schema exploration through a unified Java and TypeScript codebase with Docker-backed development workflows.